MLT Week 6

Goodness of Maximum Likelihood Estimator for Linear Regression

We observed that $w^* = \hat{w}_{ML}^{(XX^T)^+XY}$ when noise follows zero mean Gaussian distribution.

Now we want a way to understand how good $\hat{w}_{ML}$ is in estimating w.

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= E\left[\|\hat{w}_{ML}-w\|^2\right] \quad \text{(Expected value of the norm of
↪  estimator & $w\in\mathbb{R}^d$, over randomness in $y$)}
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Warning: equation could not be typeset; shown as written.

$$= \sigma^2 \cdot \text{trace}\left[(XX^T)^+\right]$$

$\rightarrow$ The goodness of the estimator is not just by the noise but by how the features are related to each other. $\rightarrow$ We cannot touch σ^2 as it is given but we def. can fool around with the features to get a better estimator with lesser mean squared error.

Cross-validation for minimizing MSE

$$\text{trace}[(XX^T)^{-1}] \quad [\text{sum of diagonal entries}] \quad \text{also } [\text{sum of eigenvalues of matrix}]$$

Let the Eigenvalues of XX^T be $\{\lambda_1, \dots, \lambda_d\}$

$\therefore$ Eigenvalues of $(XX^T)^{-1}$: $\{\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_d}\}$

Mean Squared Error $(\hat{w}_{ML})$ = $E[\|\hat{w}_{ML} - w\|^2] = \sigma^2 \sum_{i=1}^{d} \frac{1}{\lambda_i}$

For some matrix A, let eigenvalues be $\{\lambda_1, \dots, \lambda_d\}$

Then what are the eigenvalues of $(A + \lambda I)$? = $\{(A_1 + \lambda), \dots, (A_d + \lambda)\}$

$$\in \mathbb{R}_{+ve}, \qquad \in \mathbb{R}_d^{d \times d}$$

$$\text{trace}(XX^T + \lambda I)^{-1} = \left(\sum_{i=1}^{d} \frac{1}{\lambda_i + \lambda}\right)$$

We want to decrease MSE by reducing the value of trace. How do we find this $\lambda \in \mathbb{R}_+$?

By the method of cross validation



Figure 1: A horizontal line divided into two parts labeled “training set” (left) and “validation set” (right).

Training Set vs Validation Set

- Train on training set & check for errors on validation set
- Pick λ which gives least error

How to decide the best way to divide the data?

K-fold cross validation

[Diagram: A row of boxes labeled $F_1, F_2, \dots, F_k$, each representing a fold.]

- Train on folds = $\{f_1, \dots f_{i-1}, f_{i+1}, \dots, f_k\}$
- Validate on f_i
- Pick λ which gives least avg error

$$\hat{w}_{new} = (XX^T + \lambda I)^{-1} Xy$$

Bayesian Modeling for linear regression

- Need a prior on w, i.e., $P(w) \in \mathbb{R}^d$

Likelihood: $y/x \sim N(w^T x, 1)$ (for simplicity, can use σ^2 as well)

A choice of priors: $w \sim N(0, \sigma^2 I)$ — covariance matrix $\begin{bmatrix} \sigma^2 & & \\ & \ddots & \\ & & \sigma^2 \end{bmatrix}$

We know,

$$P[w/\{(x_1, y_1), \dots, (x_n, y_n)\}] \propto P[(x_1, y_1), \dots, (x_n, y_n)/w] \cdot P(w)$$

$$\propto \left[\prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}}\right] \left[\prod_{i=1}^{d} e^{-\frac{(w_i-\theta)^2}{2\sigma^2}}\right]$$

$$\propto \left[\prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}}\right] e^{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{d} w_i^2\right)} = e^{-\frac{\|w\|^2}{2\sigma^2}}$$

$$\hat{w}_{MAP} = \max_w \sum_{i=1}^{n} -\frac{(y_i - w^T x_i)^2}{2} - \frac{\|w\|^2}{2\sigma^2}$$

$$= \min_w \sum_{i=1}^{n} \frac{(y_i - w^T x_i)^2}{2} + \frac{\|w\|^2}{2\sigma^2} \quad f(w)$$

Take gradient, set it to 0 & solve for $\hat{w}_{MAP}$

$$\nabla f(w) = (XX^T)w - XY + \frac{w}{\sigma^2}$$

$$\min_w w - \left[XX^T + \frac{\sigma^2 I}{\sigma^2}\right]^{-1} XY$$

Ridge Regression

$$\hat{w}_R = \arg_w \min_{l>\text{Ridge}} \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|^2$$

Loss Regularizer